

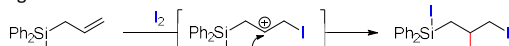
Alexie Clover, Adam Jones, and Gregory W. O'Neil.*

Department of Chemistry, WWU. *oneilg@wwwu.edu

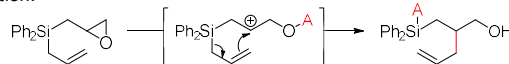
ABSTRACT: Allylsilanes can be regioselectively transformed into the corresponding 3-silylfluorohydrin in good yield using a sequence of epoxidation followed by treatment with HF-Et₃N with or without isolation of the intermediate epoxide. Various silicon substitutions are tolerated, resulting in a range of 2-fluoro-3-silylpropan-1-ol products from this method. Whereas other fluorohydrin syntheses by epoxide opening using HF-Et₃N generally require more forcing conditions (e.g., higher reaction temperature), opening of allylsilane-derived epoxides with this reagent occurs at room temperature. We attribute this rate acceleration along with the observed regioselectivity to a β-silyl effect that stabilizes a proposed cationic intermediate. The use of enantioenriched epoxides indicates that both S_N1- and S_N2-type mechanisms may be operable depending on substitution at silicon. Conformational analysis by NMR and theory along with a crystal structure obtained by X-ray diffraction points to a preference for silicon and fluorine to be proximal to one another in the products, perhaps favored due to electrostatic interactions.

Background

- Our group has developed an iodine-mediated diallylsilane rearrangement.



- We set out to investigate a similar intramolecular epoxide allylation.



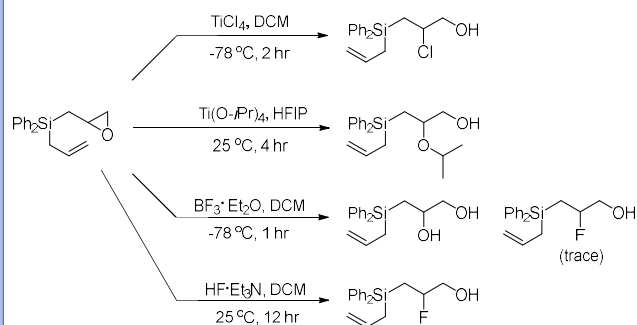
Cummins, E. J., and O'Neil, G. W. *Tetrahedron Lett.* **2017**, 58, 3406-3409.

Myers et al. *Tetrahedron Lett.* **2021**, 82, 153392.

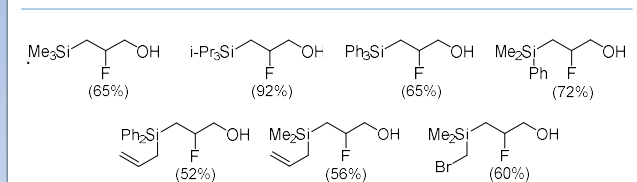
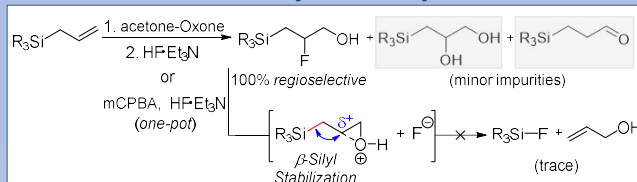
Fomina et al. *Heterocycles* **2022**, 104, DOI: 10.3987/COM-22-14735.

Initial Discovery

- Attempts at an intramolecular epoxide allylation led to trace amounts of a fluorohydrin which are valuable synthetic intermediates.



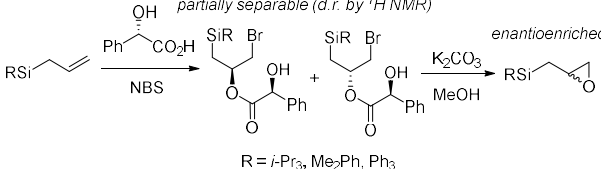
Fluorohydrin Scope



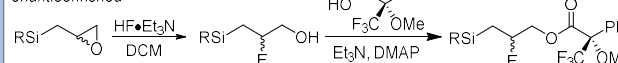
Clover, A. W., Jones, A. P., and O'Neil, G. W. *ChemRxiv. Cambridge: Cambridge Open Engage*; **2023**.

Probing the Mechanism

- Bromomandilation to enantioenriched epoxide followed by Mosher esterification.



enantioenriched

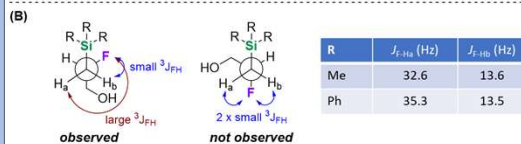
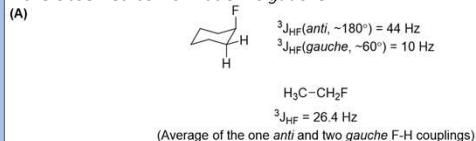


- Complete retention of ee from the *i*-Pr₃. However, an erosion of ee was observed for the other two epoxysilanes.
- Suggests S_N2 type mechanism.

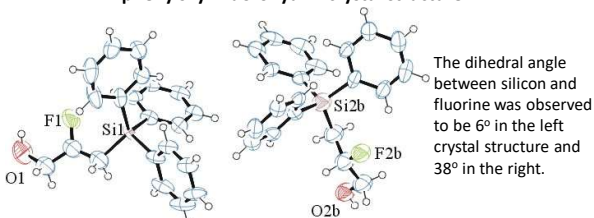
R	Epoxide ee ^a	Fluorohydrin ee ^b
<i>i</i> -Pr ₃	60	60
Me ₂ Ph	64	38
Ph ₃	59	43

A Silicon-Fluorine *gauche* Effect

- Hyperconjugation considerations would suggest a preferred anti-periplanar C-Si and C-F arrangement, with stabilization afforded by the overlap between the high energy σ C-Si and the low energy σ* C-F.
- Interestingly, ¹H NMR analysis and crystal structure provides evidence that the observed conformation is *gauche*.



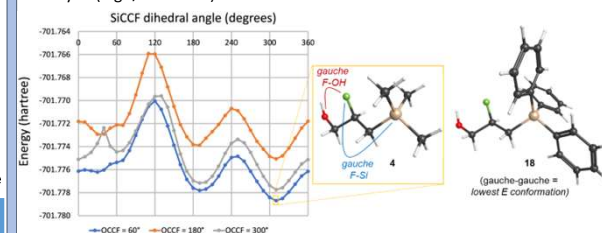
Triphenylsilyl Fluorohydrin crystal structure



The dihedral angle between silicon and fluorine was observed to be 6° in the left crystal structure and 38° in the right.

A Silicon-Fluorine *gauche* Effect

- Using both PBE and B3LYP methods, the lowest energy conformation for the trimethyl system **4** have the Si and F groups approximately *gauche* (Si-C-C-F ∅ ~ 310°) regardless of F-OH orientation (∅_{F-OH} = 60°, 180°, 300°).
- The absolute energy minimum had both Si-F and F-OH groups *gauche*, which is also true for the Triphenyl system **18** (calculated energy minimum at ∅_{Si-F} ~ 50° for ∅_{F-OH} = 60°), which is in fairly good agreement with results from X-ray analysis (e.g. ∅_{Si-F} = 38°).



Results from DFT-PBE conformational analysis of 2-fluoro-3-(trimethylsilyl)propan-1-ol (**4**). For three different F-OH orientations (∅_{F-OH} = 60°, 180°, or 300°), the lowest energy conformer contained a *gauche* Si-F arrangement (∅_{Si-F} ~ 310°).

Tamao Oxidation

- Initial attempts suffered from a competing elimination.

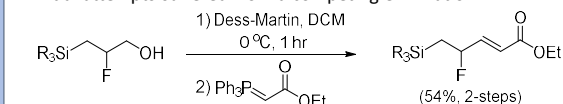


Table 1. Tamao-Fleming conditions tested.

- R = Ph₃: mCPBA, KHF₂, DMF
- R = Ph₃: TBAF, DCM, Δ
- R = Ph₃: BF₃·2AcOH, DCM, Δ
- R = PhMe₂: BF₃·2AcOH, DCM

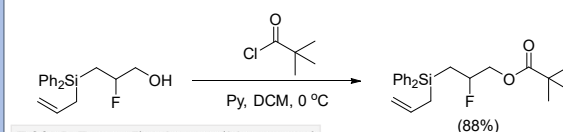
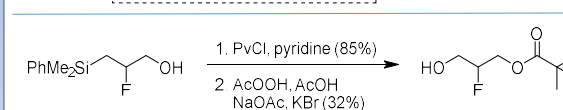
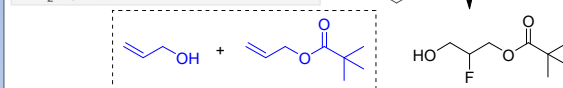


Table 2. Tamao-Fleming conditions tested.

- BF₃·2AcOH, DCM
- I₂, MeOH, DCM⁺
- I₂, CyOH, DCM⁺



Ibrahim, S. M. S.; Banerjee, K.; Slater, K. A.; Friestad, G. K. *Tetrahedron Lett.* **2017**, 58, 4864-4866.